EXPERIMENT 21: BIRD CAN FLY OR NOT

Aim

To write a **Prolog program to determine whether a particular bird can fly or not** using facts and rules, and to execute the required queries.

Algorithm

1. Start.
2. Define facts for birds that can fly.
3. Define facts for birds that cannot fly.
4. Create a can_fly rule.
5. Query the database for a particular bird.
6. Display whether the bird can fly or not.
7. Stop.

Source Code

```
% Bird Flying Database


bird(sparrow).

bird(eagle).

bird(parrot).

bird(penguin).

bird(ostrich).


can_fly(sparrow).

can_fly(eagle).

can_fly(parrot).


cannot_fly(penguin).

cannot_fly(ostrich).
```

```
fly(Bird) :-
    can_fly(Bird).


fly(Bird) :-
    cannot_fly(Bird),
    fail.
```

Queries and Output

```
?- can_fly(eagle).
true.
?- can_fly(penguin).
false.
```

Result

The Prolog program successfully determines whether a given bird can fly or not using facts and logical rules.